*1. Write the system of equations that corresponds to the following augmented matrix.

$$\left[\begin{array}{rr|r} 1 & -5 & 2 \\ -4 & 7 & -6 \end{array}\right]$$

Complete the system below.

$$\begin{cases} x - \square\, y = \square \\ \square\, x + \square\, y = \square \end{cases}$$

*2. Solve the following system of equations using matrices (row operations). If the system has no solution, say that it is inconsistent.

$$\begin{cases} 2x - 8y = -2 \\ 3x + 3y = 4 \end{cases}$$

Select the correct choice below and, if necessary, fill in the answer box(es) to complete your choice.

- **A.** The solution is (☐ , ☐).
(Simplify your answers.)
- **B.** There are infinitely many solutions. The solution can be written as
$\{(x,y) \mid x = \square$, y is any real number$\}$.
(Simplify your answer. Type an expression using y as the variable.)
- **C.** The system is inconsistent.

3. Solve the system of equations by elimination, if a solution exists.

$$\begin{cases} 12x - 11y = 16 \\ 6x - 5.5y = -18 \end{cases}$$

Select the correct answer below and, if necessary, fill in any answer boxes to complete your choice.

- **A.** The solution of the system is x = ☐ and y = ☐. (Type integers or simplified fractions.)
- **B.** There are infinitely many solutions.
- **C.** There is no solution.

4. Solve the system of equations by any convenient method, if a solution exists.

$$\begin{cases} 4x + 4y = 5 \\ x = 6y + 6 \end{cases}$$

Select the correct answer below and, if necessary, fill in any answer boxes to complete your choice.

- **A.** The solution is x = ☐ and y = ☐. (Type integers or simplified fractions.)
- **B.** There are infinitely many solutions.
- **C.** There is no solution.

*5. Solve the system using the method of your choice.

$$0.8x + 0.3y = 3.3$$
$$0.3x + 0.6y = 1.4$$

Select the correct choice below and, if necessary, fill in the answer box to complete your choice.

- **A.** The system has a single solution. The solution set is $\{\ \square\ \}$.
  (Type an ordered pair. Simplify your answer.)
- **B.** There are infinitely many solutions and the equations are dependent. The solution set is $\{(x,y)|0.8x + 0.3y = 3.3\}$.
- **C.** The system is inconsistent. The solution set is the empty set.

*6. Solve the following system of equations using matrices (row operations). If the system has no solution, say that it is inconsistent.

$$\begin{cases} x + 4y = 2 \\ 3x + 12y = 6 \end{cases}$$

Select the correct choice below and, if necessary, fill in the answer box(es) to complete your choice.

- **A.** The solution is $(\square, \square)$.
  (Simplify your answers.)
- **B.** There are infinitely many solutions. The solution can be written as $\{(x,y) \mid x = \square$, y is any real number$\}$.
  (Simplify your answer. Type an expression using y as the variable.)
- **C.** The system is inconsistent.